

3_Predicting-Plant-Height-linear-reggression-.R

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```
# PROJECT 3: Predicting Plant Height using Linear Regression  
# Goal: Building simple and multiple regression models to predict Height
```

```
# 1. Load Libraries  
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
library(readxl)
```

```
## Warning: package 'readxl' was built under R version 4.5.3
```

```
# 2. Read Data  
proj1 <- read_excel("C:/Users/prath/OneDrive/Desktop/proj1.xlsx",  
                   col_types = c("text", "numeric", "numeric",  
                                "numeric", "numeric", "text"))  
d<- proj1    #assaining object "d" for the dataset  
View(proj1)  
  
# 3. Simple Linear Regression: Length ~ Shoot_Length  
lm1 <- lm(L ~ B, data= d) #linear model for L and B  
lm2 <- lm(L ~ SL, data= d) #linear model for L and SL  
lm3 <- lm(L ~ RL, data= d) #linear model for L and RL  
s1<- summary(lm1)  
s2<- summary(lm2)  
s3<- summary(lm3)  
print(s1);print(s2);print(s3)
```

```
##
## Call:
## lm(formula = L ~ B, data = d)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.80871 -0.40594 -0.07887  0.41836  1.84698
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 39.23915    0.65786   59.65  <2e-16 ***
## B           6.03693    0.07825   77.14  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.662 on 98 degrees of freedom
## Multiple R-squared:  0.9838, Adjusted R-squared:  0.9836
## F-statistic: 5951 on 1 and 98 DF,  p-value: < 2.2e-16
```

```
##
## Call:
## lm(formula = L ~ SL, data = d)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.05012 -0.22563 -0.01415  0.19666  0.62776
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 10.706049    0.499740   21.42  <2e-16 ***
## SL          1.264860    0.007982  158.47  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3243 on 98 degrees of freedom
## Multiple R-squared:  0.9961, Adjusted R-squared:  0.9961
## F-statistic: 2.511e+04 on 1 and 98 DF,  p-value: < 2.2e-16
```

```
##
## Call:
## lm(formula = L ~ RL, data = d)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.21474 -0.19341 -0.00115  0.20905  1.53714
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 21.23348    0.54625   38.87  <2e-16 ***
## RL          2.46913    0.01964  125.75  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4082 on 98 degrees of freedom
## Multiple R-squared:  0.9938, Adjusted R-squared:  0.9938
## F-statistic: 1.581e+04 on 1 and 98 DF,  p-value: < 2.2e-16
```

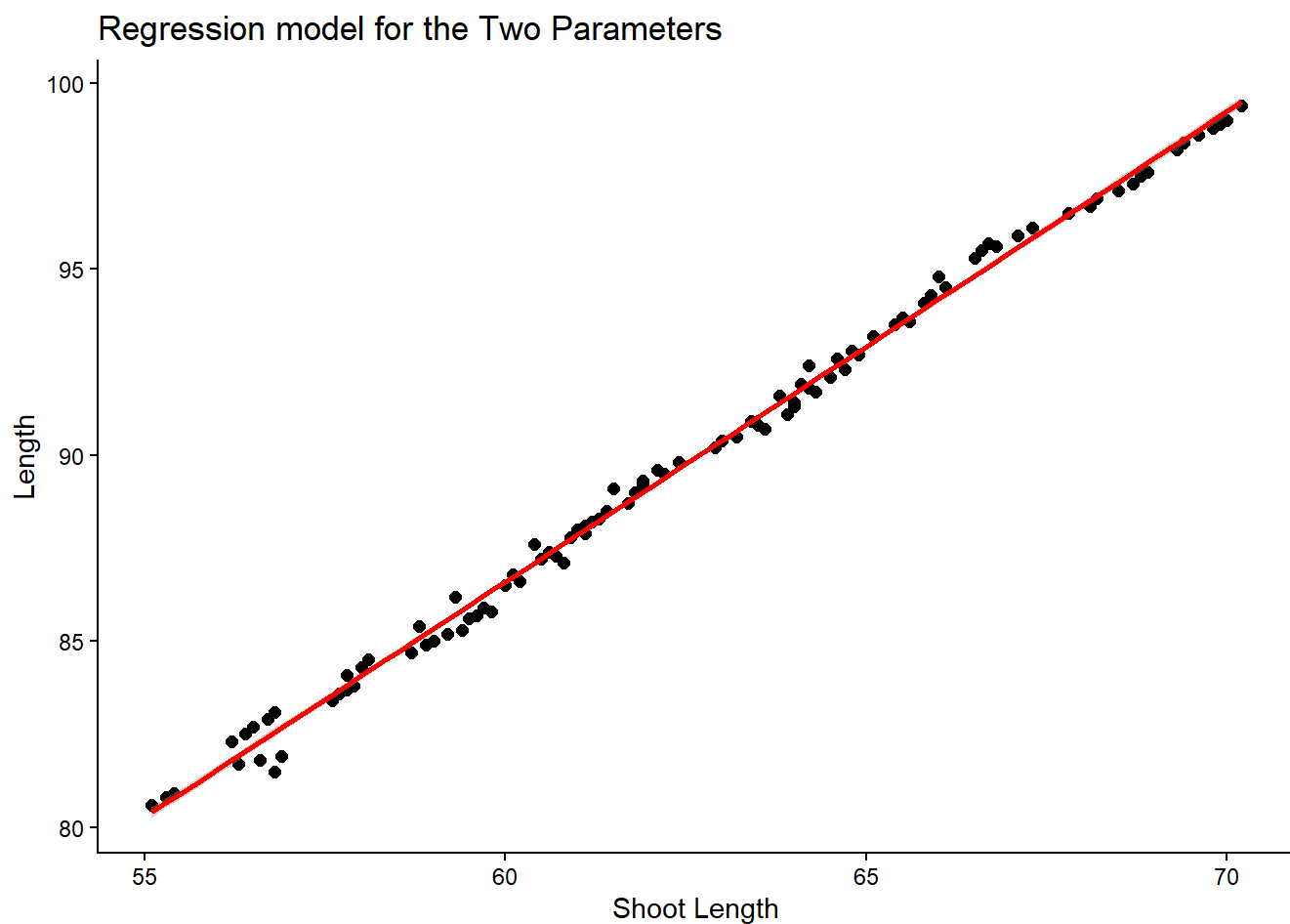
```
# 4. Multiple Linear Regression: Height ~ Shoot + Root + Width
mlr <- lm(L ~ B+SL+RL, data= d) #mlr object for multiple linear regression
ms<- summary(mlr)               #object created for summary
print(ms)
```

```
##
## Call:
## lm(formula = L ~ B + SL + RL, data = d)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.10450 -0.19780 -0.03678  0.19368  0.77232
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 13.9736    1.6671   8.382 4.39e-13 ***
## B           0.2016    0.3224   0.625  0.5332
## SL          0.9346    0.1252   7.464 3.79e-11 ***
## RL          0.5652    0.2343   2.412  0.0177 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3162 on 96 degrees of freedom
## Multiple R-squared:  0.9964, Adjusted R-squared:  0.9963
## F-statistic: 8807 on 3 and 96 DF,  p-value: < 2.2e-16
```

#5. Regression Line Visualization

```
pr<- ggplot(d, aes(SL, L))+  
  geom_point(colour = "black", size=2)+  
  geom_smooth(method = "lm", colour = "red")+  
  labs(title = "Regression model for the Two Parameters",  
        x="Shoot Length", y="Length")+  
  theme_classic()  
print(pr)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
ggsave(filename = "Regression_map.png", units = 'in', height = 8, width = 8, dpi=300)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
# 6. Model Diagnostics (Residual Check)
```

```
png(filename = "Residual_check.png", width = 480, height = 480, units = "px", pointsize = 12,  
     bg = "white", res = NA)
```

```
par(mfrow = c(2, 2))
```

```
plot(mlr)
```

```
par(mfrow= c(1,1))
```

```
dev.off()
```

```
## png
```

```
## 2
```